PROJECT BRIEFING

RSwitch Data Intelligence Platform (RDIP)

Date: 24 August 2026

Prepared by: RDIP Project Team

1. Problem Statement & Proposed Solution

The Problem

RSwitch is Rwanda's national payment switch, connecting commercial banks, e-money issuers, microfinance institutions (MFIs), and SACCOs. Today, RSwitch's operational visibility into these participating institutions is scattered across spreadsheets, emails, and disconnected systems. There is no single place to see:

- Whether a given institution's data integration is actually working, in real time
- Whether that institution is meeting its regulatory and compliance deadlines
- What incidents, risks, or unusual patterns are emerging across the network
- What governance decisions and action items are pending across the Board, committees, and working groups

Compliance deadlines are tracked manually. Integration failures are discovered late, not in real time. Governance documents live in scattered folders. There is no structured way to flag unusual participant behaviour before it becomes a problem. This creates operational risk for RSwitch and slows down the compliance and oversight work its team is responsible for.

The Proposed Solution

RDIP (RSwitch Data Intelligence Platform) is a single, role-based internal platform that consolidates all of this into one system: integration monitoring, participant compliance tracking, governance coordination, AI-assisted operational insights, and financial inclusion reporting. It replaces fragmented manual tracking with one authoritative operational view, built specifically around how RSwitch's Compliance, Business/Operations, and Technical/Admin teams actually work.

2. What Our Project Does

RDIP is a web-based platform (desktop-first, browser-accessible) built around three internal RSwitch user roles: Compliance, Business, and Technical/Admin. Each role sees a tailored dashboard and only has access to the modules relevant to their function. The platform covers:

Module	What it does
Integrations	Tracks whether each participating institution's data connection to RSwitch is healthy, and manages both manual file uploads and (over time) automated API feeds.
Pipeline Monitor	Real-time status of every data-ingestion job — running, completed, failed — with step-level logs.
Participant 360	A single consolidated profile per institution: integration status, submission history, compliance status, incidents, technical performance, and a composite risk score.

Module	What it does
Governance	A central repository for board and committee documents, meeting calendars, decisions, and action items with due-date tracking.
Compliance Monitoring	Tracks required regulatory documents per institution, deadlines, incidents, and remediation, with automated reminders and escalation.
AI Decision Support	Flags unusual patterns across participants (e.g. sudden spikes in failures) as advisory alerts for staff to investigate — it does not make automated decisions.
Financial Inclusion	Aggregated reporting on inclusion metrics segmented by institution type (bank / e-money / MFI / SACCO) and region.
Reports & Admin	On-demand and scheduled report generation (PDF/Excel), plus user management, role assignment, and audit logging.

The system is currently in active development. The team has completed Sprints 1–2 of a 12-sprint plan, with all 14 core screens built against a defined design system, and is working from a Software Requirements Specification (v1.1) that was reviewed and revised directly with RSwitch stakeholders.

3. What QT Holding Does With RSwitch

QT Holding Ltd is a Rwandan holding company that owns two subsidiaries relevant to this project:

- RSwitch Ltd — Rwanda's national electronic payment switch, operating under brands including SmartCash and eKash, providing interoperability and settlement services between banks, e-money issuers, and other financial institutions in Rwanda.
- QT Global Software Ltd — a Rwandan software development company providing technology solutions, including cybersecurity and digital transformation services, to public and private sector clients.

In short: RSwitch is the operating payments-infrastructure business, and QT Global Software is its sister technology-development company, both under the same parent, QT Holding Ltd.

RDIP was developed as our final-year project at AUCA, using RSwitch as our case-study organization. This was not a hypothetical exercise — RSwitch staff (including Raica Kwizera and Aimable Rukundo) reviewed our requirements directly, gave us structured feedback that shaped the current version of the specification, and provided real reference data (participant lists, bank codes, sample integration and compliance data) that the system is built around. This is the relationship and history our advocate should be aware of, particularly regarding any conversation about data provided by RSwitch and how it was used during development.

4. How We Plan to Help RSwitch

The platform is designed to directly address RSwitch's stated operational priorities, as confirmed in stakeholder discussions:

- Give RSwitch a single operational view of every participating institution, replacing scattered spreadsheets and email threads.
- Replace manual, fragmented data collection with structured submission workflows that can progressively automate over time — starting with manual upload, moving toward automated API feeds as institutions are ready.

- Enable proactive compliance monitoring, with automated deadline tracking and escalation instead of manual chasing.
- Provide a central governance repository for the Board, steering committees, and working groups, with reminders and accountability for action items.
- Support AI-assisted identification of unusual operational patterns and risks across participants — advisory only, never automated enforcement.
- Produce financial inclusion reports segmented by institution type and region, supporting RSwitch's broader mandate.

The team continues to build and test this in direct coordination with RSwitch feedback, with the intention of delivering a system RSwitch could put into real operational use.